

OOT2

Mathematics History and Technology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher bought a clay-tablet replica for \$35 at a museum shop. The tablet shows the Babylonian number 1;20, meaning 1 in the second place and 20 in the first. Her class meets right after lunch. What is the value in our base-10 system?

- A. 80
- B. 3,620
- C. 21
- D. 120

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

How would the base-10 number 150 be written in Babylonian base-60 notation?

- A. 15;0
- B. 2;30
- C. 1;50
- D. 30;2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A fifth-grade teacher shows the Egyptian method for 13×24 using a doubling table: $1 \rightarrow 24$, $2 \rightarrow 48$, $4 \rightarrow 96$, $8 \rightarrow 192$. She printed the table on colored card stock. Which rows should be added to get the product?

- A. $24 + 48 + 96 + 192$
- B. None; doubling only works when the multiplier is even.
- C. $24 + 96 + 192$
- D. $48 + 96 + 192$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In an Egyptian doubling table for some number times 15, a scribe added the rows for 1, 2, and 16. What was the other factor?

- A.** 3
- B.** 18
- C.** 19
- D.** 32

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

During a unit that uses a 1920s arithmetic textbook as a prop, a student asks where our digits and zero came from. The class has 24 students. Which answer is historically accurate?

- A.** Decimal place value with zero developed in India and reached Europe through Arabic-language scholars.
- B.** No culture before India used any kind of placeholder.
- C.** The Romans invented zero to simplify their numerals.
- D.** The digits were invented in Arabia, as the name “Arabic numerals” shows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A geometry teacher with a new set of compasses and 30 protractors wants students to experience how classical Greek geometry built its results. Which activity best models the axiomatic method?

- A.** Students measure the angles of many triangles and record that they add to 180° .
- B.** Students memorize a list of theorems in order.
- C.** Students prove the parallel postulate from the other postulates.
- D.** Students derive a triangle result from a short list of stated postulates, citing one for each step.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student who just finished a project on ancient Greece and earned an A says the relation $a^2 + b^2 = c^2$ was purely a Greek discovery. Which response is most accurate?

- A. Correct; no one knew it before the Greeks.
- B. The Babylonians gave the first general deductive proof.
- C. It was first discovered in Egypt, and only there.
- D. Babylonian, Chinese, and Indian mathematicians knew and used the relation; the Greek tradition is known for a general deductive proof.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8.

Using dynamic geometry software licensed for 30 laptops, students drag the vertices of a triangle 200 times. Every time, the angle sum shows 180° . The teacher is being observed that day. What have the students established?

- A. Only that the property holds for the 200 triangles shown.
- B. A proof, because 200 cases were tested.
- C. Probably a software rounding error.
- D. A strong conjecture that still needs a deductive proof.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A student drags a quadrilateral and finds one case where a claimed property fails. What does that single case establish?

- A. The claim is still true for most quadrilaterals, so it stands.
- B. The general claim is false.
- C. The software must be wrong.
- D. Nothing; one case is not enough evidence.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A student with a brand-new calculator enters $2 \div 3$, sees 0.6666667, then multiplies that display by 3 and gets 2.0000001. He argues that $(2/3) \times 3$ does not equal 2. The class is halfway through a fractions review. What is the best teacher response?

- A. Ban calculators for the rest of the unit.
- B. Explain that the display rounds 0.666... and re-entering the rounded value adds error; exactly, $(2/3) \times 3 = 2$.
- C. Tell him the calculator is defective and should be replaced.
- D. Agree, since the machine computed it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which is the most appropriate use of a computer algebra system (CAS) in a unit on factoring?

- A. Use the CAS only to draw graphs of the factored forms.
- B. Students use the CAS for every problem so they never need to learn factoring.
- C. Students factor by hand, then use the CAS to check and to explore patterns across many cases.
- D. The CAS should never be used, since it makes students dependent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher planning a quadratics unit for a room with 32 desks wants to use history to make the unit more student-centered. The school's new projector arrives next week. Which plan does that best?

- A. Students solve a 9th-century completing-the-square problem with drawn squares, then compare it with the modern algebraic method.
- B. Students take a quiz on dates of mathematical discoveries.
- C. Skip history, since tests never include it.
- D. The teacher lectures on the life of a famous mathematician.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher bought a clay-tablet replica ~~for \$35 at a museum shop~~. The tablet shows the Babylonian number 1;20, meaning 1 in the second place and 20 in the first. ~~Her class meets right after lunch~~. What is the value in our base-10 system?

Target word: convert base 60 to base 10 **Decides:** $1 \times 60 + 20$.

A	Correct	$60 + 20 = 80$.
B	Adjacent category	Puts the 1 in the 60^2 place, one place too high.
C	Dropped element	Adds the digits and drops the place value.
D	Confusion set	Reads 1;20 as the base-10 numeral 120.

2. Answer B Concept 1 · transfer

How would the base-10 number 150 be written in Babylonian base-60 notation?

Target word: convert base 10 to base 60 **Decides:** $150 = 2 \times 60 + 30$.

A	Adjacent category	Uses base-10 grouping: 15 tens.
B	Correct	Two sixties and thirty ones.
C	Confusion set	Splits by 100, not 60.
D	Inverted operation	Reverses the order of the places.

3. Answer C Concept 2

A fifth-grade teacher shows the Egyptian method for 13×24 using a doubling table: $1 \rightarrow 24$, $2 \rightarrow 48$, $4 \rightarrow 96$, $8 \rightarrow 192$. ~~She printed the table on colored card stock~~. Which rows should be added to get the product?

Target word: which rows to add **Decides:** $13 = 8 + 4 + 1$.

A	Absolute language	Adding every row gives 15×24 .
B	Adjacent category	Any whole number is a sum of powers of 2, odd ones included.
C	Correct	Rows 1, 4, and 8 sum to 13; $24 + 96 + 192 = 312 = 13 \times 24$.
D	Wrong stem number	Rows 2, 4, 8 sum to 14, not 13.

4. Answer C Concept 2 · transfer

In an Egyptian doubling table for some number times 15, a scribe added the rows for 1, 2, and 16. What was the other factor?

Target word: recover the multiplier from the rows **Decides:** Add the row labels: $1 + 2 + 16$.

A	Confusion set	Counts the rows instead of adding their labels.
B	Dropped element	Drops the row for 1.
C	Correct	$1 + 2 + 16 = 19$.
D	Inverted operation	Multiplies the labels instead of adding them.

5. Answer A Concept 3

During a unit ~~that uses a 1920s arithmetic textbook as a prop~~, a student asks where our digits and zero came from. ~~The class has 24 students.~~ Which answer is historically accurate?

Target word: origin of place value with zero **Decides:** Developed in India; carried to Europe through Arabic-language scholarship.

A	Correct	That is why the digits are called Hindu-Arabic numerals.
B	Absolute language	Late Babylonian scribes used a placeholder sign, and the Maya had a zero.
C	Confusion set	Roman numerals have no zero or place value.
D	Adjacent category	The name reflects how Europe received them, not where they began.

6. Answer D Concept 4

A geometry teacher ~~with a new set of compasses and 30 protractors~~ wants students to experience how classical Greek geometry built its results. Which activity best models the axiomatic method?

Target word: activity that models deduction from axioms **Decides:** The axiomatic method derives results from stated assumptions.

A	Confusion set	That is inductive evidence, not deduction.
B	Adjacent category	Knowing results is not deriving them.
C	Inverted operation	It cannot be proved from the others; that is why non-Euclidean geometries exist.
D	Correct	Deduction from explicit starting assumptions is the method.

7. Answer D Concept 5

A student ~~who just finished a project on ancient Greece and earned an A~~ says the relation $a^2 + b^2 = c^2$ was purely a Greek discovery. Which response is most accurate?

Target word: multicultural origin of a result **Decides:** Earlier and independent uses exist in several cultures.

A	Absolute language	Babylonian tablets list right-triangle number triples centuries earlier.
B	Inverted operation	They used the relation computationally; general proof is the Greek contribution.
C	Confusion set	Replaces one single-origin story with another.
D	Correct	Tablets and texts from these cultures show the relation in use.

8. Answer D Concept 6

Using dynamic geometry software ~~licensed for 30 laptops~~, students drag the vertices of a triangle 200 times. Every time, the angle sum shows 180° . ~~The teacher is being observed that day.~~ What have the students established?

Target word: status of a dragging result **Decides:** Many confirming cases give a conjecture, not a proof.

A	Adjacent category	True but incomplete: the evidence supports a general conjecture worth proving.
B	Absolute language	No finite number of cases proves a claim about all triangles.
C	Confusion set	Consistent results are evidence, not an error.
D	Correct	Examples, however many, do not prove a general claim.

9. Answer B Concept 6 · transfer

A student drags a quadrilateral and finds one case where a claimed property fails. What does that single case establish?

Target word: status of a counterexample **Decides:** One counterexample disproves a universal claim.

A	Adjacent category	A universal claim must hold in every case.
B	Correct	A single valid counterexample is enough.
C	Confusion set	Check the construction, but a valid counterexample stands.
D	Inverted operation	Applies the rule for proving to disproving; one failure settles it.

10. Answer B Concept 7

A student ~~with a brand new calculator~~ enters $2 \div 3$, sees 0.6666667, then multiplies that display by 3 and gets 2.0000001. He argues that $(2/3) \times 3$ does not equal 2. ~~The class is halfway through a fractions review.~~ What is the best teacher response?

Target word: interpret technology output **Decides:** The display is a rounded version of a repeating decimal.

A	Absolute language	Avoids the learning opportunity instead of using it.
B	Correct	Interpreting rounding is the teachable point.
C	Confusion set	It is working correctly; the issue is rounding.
D	Inverted operation	Treats a rounded display as exact.

11. Answer C Concept 7 · transfer

Which is the most appropriate use of a computer algebra system (CAS) in a unit on factoring?

Target word: appropriate technology role **Decides:** Technology should support and extend reasoning, not replace it.

A	Adjacent category	Graphing is a different tool role and misses checking and exploring factors.
B	Inverted operation	Replaces the understanding the unit is meant to build.
C	Correct	The CAS extends reasoning students already have.
D	Absolute language	Used well, a CAS deepens exploration.

12. Answer A Concept 8

A teacher planning a quadratics unit ~~for a room with 32 desks~~ wants to use history to make the unit more student-centered. ~~The school's new projector arrives next week.~~ Which plan does that best?

Target word: student-centered use of history **Decides:** Students engage with historical methods themselves.

A	Correct	Working the historical method builds understanding and shows mathematics as human work.
B	Confusion set	Dates are trivia, not mathematical insight.
C	Absolute language	History can deepen understanding of tested content.
D	Adjacent category	Biography is history, but it is teacher-centered.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Babylonian base-60 place value	Babylonian numerals are base 60: each place is worth 60 times the place to its right, a system still visible in minutes, seconds, and degrees.	Reading the places as powers of 10.	Numbers written in groups separated by semicolons or commas, like 1;20.
2	Egyptian doubling multiplication	Egyptian scribes multiplied by doubling and adding the rows whose multipliers sum to the other factor, which is really a binary decomposition.	Adding every row, or picking rows that do not sum to the multiplier.	A doubling table and “which rows are added?”
3	Zero and Hindu-Arabic place value	Decimal place value with a true zero developed in India and reached Europe through Arabic-language scholarship.	Crediting Arabia or Rome with inventing the system, or saying no earlier culture used a placeholder.	“Where did our number system come from?”
4	The axiomatic method	Classical Greek geometry derived every result deductively from stated definitions, postulates, and common notions.	Equating measuring many examples with proof.	An activity meant to show how a theorem follows from starting assumptions.
5	Mathematics across cultures	Many results, such as the right-triangle relation $a^2 + b^2 = c^2$, were known and used independently in Babylon, China, and India; the Greek tradition added a general deductive proof.	Treating a result as the invention of one culture or person.	A claim that a theorem is “purely” from one civilization.
6	Dynamic geometry: conjecture vs. proof	Dragging in dynamic geometry software builds strong conjectures; a proof is still needed. One counterexample disproves a general claim.	Treating many confirming screens as proof, or dismissing a single counterexample.	Students drag a figure and see a property hold or fail.
7	Interpreting technology output	Calculators and graphing tools round and window their results; the teacher must help students interpret output against exact mathematics rather than accept or reject it blindly.	Blaming a broken device, or treating the display as exact truth.	A decimal display that seems to contradict an exact result.
8	History as a student-centered tool	History enriches learning most when students work historical problems and methods themselves and compare them with modern ones.	Reducing history to names, dates, and biographies.	A teacher planning how to bring history into a unit.

Course scope (public description): In Math History and Teaching, students will learn about a variety of technological tools for doing mathematics and develop a broad understanding of the historical development of mathematics. Mathematics is a very human subject that comes from the macro-level sweep of cultural and societal change as well as the micro-level actions of individuals with personal, professional, and philosophical motivations. This course will focus on the historical development of mathematics, including contributions of significant figures

and diverse cultures. Students will learn to evaluate and apply technological tools and historical information to create an enriching student-centered mathematical learning environment.